

DEV_IMPL_GUIDE - SUB-WF-TERMINATE-01 Subscription Termination

1. Purpose

This guide explains termination implementation as the flow executes at runtime.

Use it with:

- DD_SUB-WF-TERMINATE-01-Subscription_Termination-v1.0.md
- DD_SUB-WF-TERMINATE-01-FLOW-WIK-v1.0.md
- DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md
- DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md

This guide is the parent-level implementation flow. Operator BPMN details are in the FLOW satellite.

2. Runtime Actors

Runtime component	What it does
subscription-workflow-api pod	Receives preview/terminate request, validates, starts operation.
SUB-LM module/API	Provides authoritative subscription status and mutation APIs.
SUB-WF framework	Handles idempotency, in-flight operation interruption, and Camunda start.
BIL-01	Computes TERMINATION_INTENT settlement.
BIL-04	Calls DUNNING_TERMINATION at final dunning step.
FUL-05	Revokes network/service access.
FUL-07/HomePass release	Frees HomePass slot, or approved stub until FUL-07 ships.
Drools	Validates trigger, role, cooling-off, disposition, and routing.
Camunda	Runs termination BPMN.
Worker apps	Execute BPMN service task topics.
OSR-RMA-01 EQP	Creates equipment pickup work order from termination event.
NOT-01	Sends customer/admin notification after success/failure.

3. Before The Request Can Work

Required:

- subscription_termination_config seeded.
- SUB-LM supports PENDING_TERMINATION and TERMINATED.
- SUB-LM supports termination master fields.
- subscription_operation_config has (operator_code=WIK, operation_kind=TERMINATE) -> sub-terminate-wik.
- Drools package rules.subscription.terminate.wik.
- BPMN process sub-terminate-wik.
- Worker apps subscribe to required topics.
- BIL-01 supports TERMINATION_INTENT.
- FUL-05 can revoke service.
- HomePass release worker is implemented or approved stub is in place.

4. Step 1 - Caller Uses Preview

Preview is optional but recommended for customer/call-center termination. It lets the caller show settlement and equipment handling before final confirmation.

Endpoint:

```
POST /api/subscriptions/sub_01HZ_KAMAU_FIBER/terminate/preview HTTP/1.1
Authorization: Bearer <jwt-or-service-token>
Content-Type: application/json

{
  "operatorCode": "WIK",
  "terminationTrigger": "CUSTOMER_REQUESTED",
  "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
  "equipmentDisposition": "PENDING_COLLECTION"
}
```

The preview API:

1. Loads subscription summary from SUB-LM.
2. Loads termination config.
3. Loads equipment bindings from OSR/asset inventory approved API.
4. Calls Drools to compute cooling-off and disposition.
5. Calls BIL-01 in preview/read-only mode for TERMINATION_INTENT.
6. Returns preview response.

Do not create subscription_operation. Do not start Camunda. Do not change subscription state.

5. Step 2 - Caller Sends Terminate Request

Public endpoint:

```
POST /api/subscriptions/sub_01HZ_KAMAU_FIBER/terminate HTTP/1.1
Authorization: Bearer <jwt-or-service-token>
Idempotency-Key: term-sub_01HZ_KAMAU-20260720T143000-cc007
Content-Type: application/json

{
  "operatorCode": "WIK",
  "terminationTrigger": "CUSTOMER_REQUESTED",
  "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
  "equipmentDisposition": "PENDING_COLLECTION",
  "customerNotes": "Moving to a non-covered area; no service available at new address",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-07-20-4421"
}
```

Actor user/role comes from auth context, not request body.

6. Step 3 - API Validates Shape And Idempotency

Validate request shape:

Field	Validation
operatorCode	Required.
terminationTrigger	One of CUSTOMER_REQUESTED, ADMIN_DECISION, DUNNING_TERMINATION, CONTRACT_EXPIRED.
terminationReasonCode	Required.
equipmentDisposition	Optional; if supplied must be CUSTOMER_RETAINS, PENDING_COLLECTION, or WRITTEN_OFF.
customerNotes	Optional free text.

Idempotency behavior:

- Same key and same request hash returns existing operation response.
- Same key and different request hash returns 409 IDEMPOTENCY_KEY_REUSED.
- Check existing key and hash before inserting a new operation.
- Use a DB unique constraint or transaction guard so concurrent duplicate calls cannot create two operations.

7. Step 4 - API Loads Required Data

7.1 Subscription Summary

Load from SUB-LM source of truth:

Same app:

```
SubscriptionReadService.getSummary(subscriptionId)
```

Split service:

```
GET /internal/subscriptions/sub_01HZ_KAMAU_FIBER/summary
Authorization: Bearer <subscription-workflow-service-token>
```

Expected:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operatorCode": "WIK",
  "customerId": "cust_01HZ_KAMAU",
  "statusCode": "ACTIVE",
  "homepassId": "hp_01HZ_KAREN_017",
  "packageRef": "pkg_FIBER_HOME_50M",
  "activatedAt": "2024-01-20T10:00:00Z",
  "contractStartedAt": "2024-01-20T10:00:00Z",
  "contractEndAt": "2026-01-20T10:00:00Z",
  "contractMinimumTermMonths": 12
}
```

Do not use cache for current subscription status.

7.2 Termination Config

```
SELECT *
FROM subscription_termination_config
WHERE operator_code = 'WIK';
```

Expected:

```
{
  "operatorCode": "WIK",
  "coolingOffWindowDays": 7,
  "defaultEquipmentDisposition": "PENDING_COLLECTION",
  "dunningEquipmentDisposition": "PENDING_COLLECTION",
  "dunningAutoWriteOffThreshold": 10000.00,
  "dunningAutoWriteOffCurrency": "KES",
  "customerSelfServiceEnabled": true,
  "coolingOffSupervisorOverride": true,
  "churnSaveAttemptEnabled": false
}
```

7.3 Equipment Bindings

Load equipment bindings through OSR/asset inventory approved API or client wrapper.

Example internal API:

```
GET /internal/subscriptions/sub_01HZ_KAMAU_FIBER/equipment-bindings
Authorization: Bearer <subscription-workflow-service-token>
```

Expected:

```
[
  {
    "equipmentRequirement": "GPON_ONT",
    "equipmentRef": "ont_01HZ_KAMAU"
  },
  {
    "equipmentRequirement": "ATA",
    "equipmentRef": "ata_01HZ_KAMAU"
  }
]
```

The termination flow should not query OSR database tables directly.

8. Step 5 - API Applies Fast Rejections

Before calling the framework:

Check	Failure
Subscription is already TERMINATED.	409 ALREADY_TERMINATED
Actor role matches trigger.	403 ROLE_NOT_AUTHORIZED_FOR_TRIGGER
Customer self-service enabled if role is CUSTOMER_SELF_SERVICE.	403 CUSTOMER_SELF_SERVICE_NOT_ENABLED
Equipment disposition is valid.	400 UNKNOWN_EQUIPMENT_DISPOSITION
Termination reason exists in operator taxonomy if configured.	400 UNKNOWN_TERMINATION_REASON

Termination may proceed when another operation is in-flight, but only after interruption succeeds.

9. Step 6 - API Handles In-Flight Operation Interruption

If another state-changing operation is in-flight for the subscription:

1. Framework sends message `external-cancel-requested` to the active BPMN.
2. Framework waits up to 60 seconds for cancellation acknowledgement.
3. Existing operation runs its compensation path and finalizes as `CANCELLED` or configured interrupted state.
4. Termination proceeds.

If the active operation does not acknowledge cancellation:

```
409 Conflict
{
  "errorCode": "OPERATION_INTERRUPTION_FAILED",
  "activeOperationId": "subop_01HZ_KAMAU_PAUSE_PRIOR"
}
```

10. Step 7 - API Calls Drools

Package:

```
rules.subscription.terminate.wik
```

Facts:

```
{
  "operatorCode": "WIK",
  "subscription": {
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "statusCode": "ACTIVE",
    "activatedAt": "2024-01-20T10:00:00Z",
    "contractMinimumTermMonths": 12
  },
  "request": {
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "equipmentDisposition": "PENDING_COLLECTION"
  },
  "config": {
    "coolingOffWindowDays": 7,
    "defaultEquipmentDisposition": "PENDING_COLLECTION",
    "dunningEquipmentDisposition": "PENDING_COLLECTION",
    "customerSelfServiceEnabled": true,
    "coolingOffSupervisorOverride": true,
    "churnSaveAttemptEnabled": false
  },
  "actor": {
    "userId": "cc-agent-wik-007",
    "role": "CALL_CENTER"
  },
  "runtime": {
    "now": "2026-07-20T14:30:00Z"
  }
}
```

Expected decision:

```
{
  "eligible": true,
  "resolvedEquipmentDisposition": "PENDING_COLLECTION",
  "coolingOffApplicable": false,
  "requiresCoolingOffSupervisorDecision": false,
  "requiresChurnSaveAttempt": false
}
```

```
}
```

11. Drools Rules To Configure

11.1 Already Terminated

```
rule "terminate rejects already terminated subscription"
when
    $req: TerminateRequestFact()
    $sub: SubscriptionStateFact(subscriptionId == $req.subscriptionId, statusCode == "TERMINATED")
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ALREADY_TERMINATED")
    }
end
```

11.2 Trigger Role Gating

```
rule "customer requested termination allowed roles"
when
    $req: TerminateRequestFact(
        terminationTrigger == "CUSTOMER_REQUESTED",
        eligible == null,
        initiatingActorRole not in ("CUSTOMER_SELF_SERVICE", "CALL_CENTER", "SUBSCRIPTION_OPERATOR")
    )
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ROLE_NOT_AUTHORIZED_FOR_TRIGGER")
    }
end

rule "admin decision termination requires admin"
when
    $req: TerminateRequestFact(
        terminationTrigger == "ADMIN_DECISION",
        eligible == null,
        initiatingActorRole != "SUBSCRIPTION_ADMIN"
    )
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ROLE_NOT_AUTHORIZED_FOR_TRIGGER")
    }
end

rule "dunning termination requires billing internal"
when
    $req: TerminateRequestFact(
        terminationTrigger == "DUNNING_TERMINATION",
        eligible == null,
        initiatingActorRole != "BILLING_INTERNAL"
    )
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ROLE_NOT_AUTHORIZED_FOR_TRIGGER")
    }
end
```

11.3 Equipment Disposition Resolution

```
rule "caller supplied equipment disposition"
when
    $req: TerminateRequestFact(equipmentDisposition != null, resolvedEquipmentDisposition == null)
then
    modify($req) {
        setResolvedEquipmentDisposition($req.getEquipmentDisposition())
    }
end

rule "default dunning equipment disposition"
when
    $req: TerminateRequestFact(
        terminationTrigger == "DUNNING_TERMINATION",
        equipmentDisposition == null,
        resolvedEquipmentDisposition == null
    )
    $cfg: TerminationConfigFact()
then
end
```

```

then
  modify($req) {
    setResolvedEquipmentDisposition($cfg.getDunningEquipmentDisposition())
  }
end

rule "default non dunning equipment disposition"
when
  $req: TerminateRequestFact(
    terminationTrigger != "DUNNING_TERMINATION",
    equipmentDisposition == null,
    resolvedEquipmentDisposition == null
  )
  $cfg: TerminationConfigFact()
then
  modify($req) {
    setResolvedEquipmentDisposition($cfg.getDefaultEquipmentDisposition())
  }
end

```

11.4 Cooling-Off

```

rule "customer termination inside cooling off"
when
  $req: TerminateRequestFact(terminationTrigger == "CUSTOMER_REQUESTED")
  $sub: SubscriptionStateFact(activatedAt != null)
  $cfg: TerminationConfigFact()
  eval(java.time.Instant.parse($sub.getActivatedAt())
    .plusSeconds($cfg.getCoolingOffWindowDays() * 86400L)
    .isAfter(java.time.Instant.parse($req.getRuntimeNow())))
then
  modify($req) {
    setCoolingOffApplicable(true)
  }
end

rule "cooling off supervisor decision required"
when
  $req: TerminateRequestFact(coolingOffApplicable == true)
  $cfg: TerminationConfigFact(coolingOffSupervisorOverride == true)
then
  modify($req) {
    setRequiresCoolingOffSupervisorDecision(true)
  }
end

```

12. Step 8 - API Starts Operation

After API validation, interruption handling, and Drools approval, the terminate API asks the shared SUB-WF framework to create the operation and start Camunda.

Same deployable:

```
OperationStartService.start(StartOperationCommand command)
```

Split service:

```

POST /internal/subscription-operations/start
Authorization: Bearer <subscription-workflow-service-token>
Content-Type: application/json

```

The payload below is the internal command passed to the framework start contract. It is not the public terminate API request.

```

{
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operationKind": "TERMINATE",
  "idempotencyKey": "term-sub_01HZ_KAMAU-20260720T143000-cc007",
  "initiatingActorUserId": "cc-agent-wik-007",
  "initiatingActorRole": "CALL_CENTER",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-07-20-4421",
  "bpmnVariables": {
    "customerId": "cust_01HZ_KAMAU",
    "homepassId": "hp_01HZ_KAREN_017",
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "resolvedEquipmentDisposition": "PENDING_COLLECTION",
    "coolingOffApplicable": false,
    "requiresCoolingOffSupervisorDecision": false,
    "requiresChurnSaveAttempt": false,
  }
}

```

```

    "equipmentBindings": [
      {
        "equipmentRequirement": "GPON_ONT",
        "equipmentRef": "ont_01HZ_KAMAU"
      }
    ],
    "customerNotes": "Moving to a non-covered area; no service available at new address"
  }
}

```

Framework starts Camunda process:

```
sub-terminate-wik
```

13. Step 9 - API Returns Accepted

```

{
  "operationId": "subop_01HZ_KAMAU_TERM",
  "operationKind": "TERMINATE",
  "bpmnProcessKey": "sub-terminate-wik",
  "currentState": "INITIATED"
}

```

14. Step 10 - BPMN Executes Worker Topics

Order	BPMN service task does this	Worker topic/message
1	Re-check policy.	drools-decide
2	Optional churn-save or supervisor decision.	User task, if configured
3	Put subscription in pending termination.	sub-lm-put-pending-status
4	Compute cycle window.	sub-lm-compute-cycle-window
5	Ask billing for settlement.	bil01-emit-intent
6	Revoke service.	ful-call-terminate
7	Wait for fulfillment result.	fulfillment-completed or fulfillment-failed
8	Release HomePass.	ful-call-release-homepass
9	Write termination master fields.	sub-lm-put-master-fields
10	Commit subscription terminated.	sub-lm-commit-state
11	Emit termination event.	kafka-emit
12	Notify customer.	notification-send
13	Finalize operation.	subscription-operation-finalize

These names are worker topics or BPMN message names, not shell commands.

15. Step 11 - Billing Settlement Details

Service task topic:

```
bil01-emit-intent
```

Worker input:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intentKind": "TERMINATION_INTENT",
  "intentContext": {
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "priorStatus": "ACTIVE",
    "coolingOffApplicable": false,
    "daysActiveInContract": 920,
    "contractMinimumTermMonths": 12,
    "equipmentDisposition": "PENDING_COLLECTION"
  },
  "cycleWindow": {
    "cycleStart": "2026-07-01",
    "cycleEnd": "2026-07-31"
  }
}

```

Expected output:

```
{
  "intentDecisionId": "bidec_01HZ_KAMAU_TERM",
  "settlement": {
    "lineItems": [
      {
        "kind": "PRORATED_REFUND",
        "amount": -1200.00,
        "currency": "KES"
      }
    ],
    "totals": {
      "net": -1200.00,
      "currency": "KES"
    },
    "refundAmount": 1200.00,
    "writeOffMarker": false
  }
}
```

SUB-WF stores this decision in BPMN variables and event payload. It does not recalculate settlement.

16. Step 12 - FUL-05 Revokes Service

Service task topic:

ful-call-terminate

Worker input:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intent": "TERMINATE_SERVICE",
  "reasonCategory": "CUSTOMER_MOVED_NO_COVERAGE",
  "equipmentDisposition": "PENDING_COLLECTION",
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_KAMAU"
    }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_TERM:terminate"
}
```

BPMN waits for:

fulfillment-completed
fulfillment-failed

Correlate on:

fulfillmentCorrelationId = subop_01HZ_KAMAU_TERM:terminate

17. Step 13 - HomePass Release

Service task topic:

ful-call-release-homepass

Worker input:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "homepassId": "hp_01HZ_KAREN_017",
  "releaseReason": "SUBSCRIPTION_TERMINATED",
  "terminationOperationId": "subop_01HZ_KAMAU_TERM"
}
```

Until FUL-07 ships, this worker may call an approved stub. The stub must be observable and must not silently drop release requests.

18. Step 14 - BPMN Writes Master Fields

Service task topic:

sub-lm-put-master-fields

Worker input:


```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "fieldsJson": {
    "terminated_at": "2026-07-20T14:30:18Z",
    "termination_trigger": "CUSTOMER_REQUESTED",
    "termination_reason_code": "CUSTOMER_MOVED_NO_COVERAGE",
    "prior_status_at_termination": "ACTIVE",
    "last_status_changed_at": "2026-07-20T14:30:18Z"
  }
}
```

This step runs before the final TERMINATED commit. If master-field writing or the final commit repeatedly fails after service revoke, route to manual recovery.

19. Step 15 - BPMN Commits TERMINATED

Service task topic:

```
sub-lm-commit-state
```

Worker input:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "expectedFromStatus": "PENDING_TERMINATION",
  "toStatus": "TERMINATED"
}
```

Only after this succeeds can the BPMN emit SubscriptionTerminated.

20. Step 16 - Success Event Is Emitted

Event:

```
SubscriptionTerminated
```

Topic:

```
sophix.subscription.subscription.terminated
```

Emitted only after SUB-LM commits TERMINATED.

OSR-RMA-01 consumes this event and creates a pickup work order when:

```
equipmentDisposition == PENDING_COLLECTION
AND equipmentBindings is not empty
```

21. Compensation Flows

Failure	BPMN compensation
Validation reject before pending	kafka-emit SubscriptionTerminationRejected, finalize failed.
Billing settlement failure	sub-lm-revert-to-prior, emit rejected, finalize failed.
FUL-05 failed before network revoke	sub-lm-revert-to-prior, emit rejected, finalize failed.
FUL-05 partial revoke or unknown state	Manual recovery.
HomePass release failed	Manual recovery or configured retry; do not commit terminal state until resolved.
Master fields/commit failed after network revoke	Manual recovery.
Kafka emit failed	Outbox retry.
Notification failed	Log and continue.

22. Manual Smoke Test

1. Seed subscription_termination_config.
2. Seed operation config (WIK, TERMINATE) -> sub-terminate-wik.
3. Create active WIK subscription.
4. Add equipment bindings for the subscription.
5. Call terminate preview and confirm no state change.

6. Send terminate API request with CUSTOMER_REQUESTED.
7. Confirm operation row exists.
8. Confirm BPMN process exists.
9. Confirm subscription moves to PENDING_TERMINATION.
10. Confirm BIL-01 receives TERMINATION_INTENT.
11. Correlate FUL-05 fulfillment-completed.
12. Confirm HomePass release worker ran.
13. Confirm subscription becomes TERMINATED.
14. Confirm termination master fields are populated.
15. Confirm outbox contains SubscriptionTerminated.
16. Confirm OSR-RMA consumes event and creates pickup work order when equipment is pending collection.
17. Confirm operation final state is COMPLETED.

23. Common Mistakes

- Reading current subscription status from cache.
- Treating termination as reversible.
- Starting termination without handling in-flight operations.
- Putting billing formulas in SUB-WF instead of using BIL-01.
- Ignoring equipment disposition in event payload.
- Committing TERMINATED before FUL-05 revokes service.
- Emitting SubscriptionTerminated before sub-lm-commit-state.
- Treating worker-topic names as shell commands.
- Failing the whole operation because NOT-01 notification failed.